

1. Administrative & Study Identification

Full Study Title: A Study to Investigate the Effect of AZD6234, AZD9550, and a Combination of AZD9550 and AZD6234 on Pharmacokinetics of Combined Oral Contraceptive Ethinyl Estradiol/Levonorgestrel in Healthy Female Participants Living With Overweight or Obesity **Short Title/Acronym:** AZD6234/AZD9550 and Oral Contraceptive PK Study **Protocol Number:** Internal Sponsor ID **NCT Number:** NCT07013643 **Posting ID:** 461770279439263604 **Sponsor/Company Name:** AstraZeneca **Investigational Product:** AZD6234, AZD9550, Ethinyl Estradiol/Levonorgestrel (EE/LEVO), and Acetaminophen (APAP) / Paracetamol (ATC Code: N02BE01) **Phase of Development:** Phase I **Trial Status:** Recruiting (Currently recruiting participants) **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To investigate the effect of AZD6234, AZD9550, and a combination of both on the pharmacokinetics (PK), safety, and tolerability of a combined oral contraceptive (CoC), ethinyl estradiol/levonorgestrel (EE/LEVO). **Secondary Objectives:** To assess the pharmacokinetics of AZD6234, AZD9550, and APAP, and to evaluate overall safety, tolerability, anti-drug antibody (ADA) development, and changes in body weight or BMI.

2.2 Background & Rationale To address the potential for drug-drug interactions (DDIs) between novel weight-management therapies (AZD6234 and AZD9550) and commonly used combined oral contraceptives (EE/LEVO). Understanding this PK interaction is a medical necessity to ensure that oral contraceptives retain their efficacy in females of childbearing potential living with overweight or obesity while undergoing these metabolic treatments.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-sequence, multiple-cohort (baseline serves as control) **Blinding:** Open-label **Randomization:** Non-Randomized (Sequential Assignment)

3.2 Planned Enrollment & Duration Target **\$N\$:** 50 targeted participants **Population Group:** Healthy female participants (Healthy Volunteers) **Number of Sites:** Multiple study sites

(Clinical Units) **Estimated Study Start:** 2024 **Estimated Primary Completion:** To be determined

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Healthy females of childbearing and non-childbearing potential. (Females of non-childbearing potential must be confirmed at the Screening Visit). 2. Body Mass Index (BMI) between 25 and 40 kg/m² (inclusive) and weigh at least 60 kg. 3. Negative pregnancy test at the Screening Visit and on admission; females of childbearing potential must not be lactating, and if heterosexually active, must agree to use an approved method of highly effective, non-hormonal contraception.

4.2 Exclusion Criteria 1. History of clinically important disease or disorder including gastroparesis, deep vein thrombosis (DVT), venous thromboembolism (VTE), cardiovascular disease, neuromuscular/neurogenic disease, severe vitamin D deficiency, type I or II diabetes mellitus, and previous surgery of the upper GI tract. 2. Glycated hemoglobin (HbA1c) $\geq$ 6.5% at screening; history of neoplastic disease; basal calcitonin >50 ng/L at screening (Cohort 2); history of acute/chronic pancreatitis or pancreatic amylase/lipase >2xULN at screening (Cohort 2); prior history of cholecystectomy or untreated cholelithiasis; personal/family history of medullary thyroid cancer (MTC) or multiple endocrine neoplasia type 2 (MEN2) (Cohort 2). 3. Any positive result for Hepatitis B surface antigen (HBsAg), Hepatitis B core antibody (HBcAb), or Human Immunodeficiency Virus (HIV). 4. History or presence of GI, hepatic, or renal disease interfering with drug absorption, distribution, metabolism, or excretion. 5. Abnormal vital signs, or clinically important abnormalities in rhythm, conduction, or morphology of the resting 12-lead ECG. 6. Current smokers or those who have smoked or used nicotine products. 7. Known or suspected history of alcohol/drug abuse; severe or ongoing clinically important allergy/hypersensitivity. 8. Statin treatment within 4 weeks prior to the start of study treatment. 9. Current use of estrogen-containing products or hormonal contraceptives (to avoid confounding the CoC assessment).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Multiple doses of AZD6234 and a combination of AZD9550 and AZD6234, administered across 2 cohorts. Each cohort consists of 5 periods: Screening, Start, Up-titration, Maintenance, and Follow-up. CoC (EE/LEVO) and APAP (Paracetamol) are administered as single assessment doses. **Route of Administration:** Subcutaneous injection (AZD6234, AZD9550), Oral tablet (EE/LEVO), Oral solution/tablet (APAP/Paracetamol). **Duration of Treatment:** Treatment exposure lasts across the Start, Up-titration, and Maintenance periods.

5.2 Concomitant & Prohibited Therapy Permitted: Approved non-hormonal highly effective contraception methods. **Prohibited:** Hormonal contraceptives, estrogen-containing birth control, and statins (prior to and during the assessment).

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Pre-study Informed Consent, Medical History, ECG, Pregnancy Test, BMI assessment, Viral screening (HIV, Hepatitis), HbA1c, calcitonin
Start Period	Period 2	Admission to Clinical Unit, baseline PK, initial dosing	Up-titration & Maintenance Periods 3 & 4 Multiple dosing of AZD therapies, PK blood sampling, EE/LEVO and APAP administration, Safety tracking
End of Study / Follow-up	Period 5	Final PK/PD assessments, Vital Signs, Exit Interview, AE review	

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints Definition: Pharmacokinetics (PK) of combined oral contraceptive components, ethinyl estradiol (EE) and levonorgestrel (LEVO), evaluated specifically by: * Area under the concentration-time curve from time 0 to infinity (AUC_{∞}) * Area under the concentration-time curve from time of dosing to the last measurable concentration (AUC_{last}) * Maximum plasma concentration (C_{max}) **Measurement Method:** Serial blood sampling and central laboratory plasma concentration analysis.

7.2 Secondary & Exploratory Endpoints 1. Number of participants with adverse events (AEs). 2. Number of participants developing detectable anti-drug antibodies (ADAs) against AZD6234 and AZD9550. 3. Area under plasma concentration-time curve from time 0 to 168 hours postdose (AUC_{0-168h}) of AZD6234. 4. Additional PK parameters for AZD6234, AZD9550, and APAP/Paracetamol. 5. Pharmacodynamic (PD) changes in body weight and BMI.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring by Clinical Unit staff (vital sign checks, ECGs, routine labs, ADA tracking) throughout the inpatient periods. **Serious Adverse Events (SAEs):** Standard protocol; reported within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Supervised internally by the Sponsor's (AstraZeneca) safety review board.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants who receive at least one dose) and PK Set (all evaluable participants with sufficient concentration data). **Sample Size Justification:** Target enrollment of 50 participants, powered according to standard Phase I PK equivalence/interaction parameters. **Handling of Missing Data:** Standard non-compartmental PK analysis rules (e.g., concentration values below the limit of quantification handled per standard protocols).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Direct onsite monitoring during Clinical Unit admission periods to verify source data and protocol adherence. **Audit Trail:** Validated electronic data capture (EDC) system with full audit trails for data clarification forms (DCFs).

11. Study Identifiers & Governance

Primary ID: NCT07013643 **Registry Number:** NCT07013643 **Posting ID:** 461770279439263604 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** PK of EE/LEVO with AZD6234 and AZD9550 in Overweight/Obese Females **Official Regulatory Title:** An Open-label, Single-sequence Multiple Cohort Study to Assess the Effect of Multiple Doses of AZD6234 and a Combination of AZD9550 and AZD6234 on the Pharmacokinetics of Single Doses of Combined Oral Contraceptive Ethinyl Estradiol/Levonorgestrel in Healthy Female Participants Living With Overweight or Obesity

12. Clinical Framework (Population & Methodology Specifics)

Disease Areas / Target Population: Healthy Participants / Healthy Volunteers **Population Classification:** Population Group **Associated Terminology Codes:** MeSH: D064368 | SNOMED CT: 106207008 **Diagnosis Threshold:** BMI between 25 and 40 kg/m^2 **Medical Methodology:** Pharmacokinetics Evaluation / DDI Assessment (Preferred UMLS Name: Pharmacokinetics) **Optimization Target:** Ensuring stable systemic exposure to CoCs alongside investigational metabolic therapies

13. Study Procedures & Methodology

Management Pathway: Inpatient Phase I Clinical Unit monitoring **Education Protocol (HCP):** Site Initiation Visits (SIV) and strict

training on PK sampling windows **Education Protocol (Patient):** Informed Consent, clinic orientation, strict adherence to non-hormonal contraception **Quality Audit Mechanism:** Real-time Source Data Verification (SDV) for PK blood draws

14. Observation & Follow-Up Schedule

Total Duration: Length of the 5 pre-defined study periods (Screening to Follow-up) **Onsite Visit Cadence:** Confinement to Clinical Unit during the Start, Up-titration, and Maintenance periods **Remote Monitoring Frequency:** N/A (Predominantly inpatient clinical monitoring) **Checklist Review Interval:** Continuous AE and vital sign monitoring during facility admission

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				